Ysabel Wang

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ACADEMIC BACKGROUND

MIT-WHOI Joint Program in Oceanography — **PhD in Physical Oceanography**
Massachusetts Institute of Technology, Cambridge MA
Woods Hole Oceanographic Institution, Falmouth MA
(2023 - present)

Texas A&M University College Station — **MS in Oceanography**
College Station TX
(2021 - 2023)

University of the Philippines Diliman — **BS in Physics**
Quezon City, Philippines
(2013 - 2018)

RESEARCH / EMPLOYMENT EXPERIENCE

MIT-WHOI Joint Program — **Graduate Research Assistant**
Advised by Dr. Magdalena Andres
(Sept 2023 – present)

Current and previous projects:

- New England Seamounts Acoustics Experiment
  - Investigated and evaluated the role of Gulf Stream warm core rings in driving cross-scale oceanic processes and their recent cumulative and instantaneous impact on physical variability of the shelf-slope environment
  - Used satellite datasets (sea surface height, temperature, and chlorophyll) to assess the evolution of warm core rings and identify shelf streamers and estimate their widths; integrated *in* situ ship-based observations to characterize a warm core ring interaction with a cyclone
  - Modified an idealized ROMS model to examine the impact of baroclinicity on streamer dynamics
- SWOT Adopt a Crossover Field Campaign – Cape Hatteras (MAB-SWOT)
  - Participated in a research cruise to retrieve an deploy Current and Pressure Equipped Inverted Echo Sounders (CPIESs) and collect hydrographic data in the Mid Atlantic Bight (MAB) (June 2024)
  - Processed and analyzed CPIES records; published a publicly available L3 dataset and technical report (see below)
- National Oceanographic Partnership Program Global Internal Waves Study
  - Identify the internal wave signal in uncorrected SWOT measurements during the MAB-SWOT Calibration and Validation (Cal/Val) phase of the SWOT deployment
  - Analyze the concurrent CPIES records from the MAB-SWOT project to identify internal wave signals
  - Compare the ground truth internal wave signals from *in situ* measurements with the uncorrected SWOT data
  - Evaluate the internal wave corrections performed on SWOT using the High Resolution Empirical Tides Model

Texas A&M University /Geochemical and Environmental Group — **Graduate Research Assistant**
Advised by Dr. Steven DiMarco
(Aug 2021 – Jul 2023)

MS Thesis: *Physical mechanisms behind the formation and fate of a coastal dense surface water mass during the 2021 Texas winter storms*

Projects:

- Ocean Acidification on a Crossroad: Enhanced Respiration, Upwelling, Increasing Atmospheric CO2, and their interactions in the northwestern Gulf of Mexico; part of the NOAA Ocean Acidification Program
  - Processed and analyzed oceanographic data on the Texas-Louisiana shelf to assess the impact of the 2021 winter events on hydrography, circulation, and carbonate chemistry
  - Supported in the hardware & software integration and deployment of a Liquid Robotics SV3 wave glider
  - Participated in field surveys in the northern Gulf of Mexico (October 2022, December 2022, March 2023)

The Marine Science Institute, University of the Philippines — **University Research Associate II**
Supervised by Dr. Cesar Villanoy and Dr. Aletta Yniguez
(Sept 2018 – Mar 2021)

Projects:

- Hazard Detection and Mitigation Tools for Opportunistic Algal Blooms in a Changing Environment
  - Conducted field surveys and participated in research cruises in the Samar and Visayan Seas
  - Developed a 3-D hydrodynamic model on Delft3D and assimilated *in situ* observations to assess the physical dynamics of harmful algal blooms in the region
- Deploying Unmanned Research Vessels to Advance Marine and Environmental Health Monitoring and Data Collection in the Philippines

- Conducted field surveys and participated in research cruises to deploy sensors and to assess the hydrodynamics and water quality in Bolinao, Pangasinan and Boracay, Aklan
- Piloted the Liquid Robotics SV3 v2 Wave Glider in Tubbataha Reefs Natural Park, Palawan

- **Kuroshio Current Observing System of the Philippines**
  - Assisted in the installation of the HF radar system for surface current detection in Gonzaga, Cagayan

I also assisted in the development of the data acquisition system of the research vessel R/V Kasarinlan that reported real-time ADCP, GPS, and weather data, and in the development of a low-cost underway data collection system.

Select Publications and Technical Reports

Wang, Y.D., Andres, M., Gawarkiewicz, G. G., Zhang, W. G. (2026). The impact of Gulf Stream warm core rings on shelf water export in the Northwestern Atlantic. Journal of Geophysical Research: Oceans (in prep).

Wang, Y.D., DiMarco, S.F. (2026). The 2021 Texas winter storms: evidence of the formation and fate of a coastal dense water mass. Geophysical Research Letters (in prep).

Wang, Y. D. & Andres, M. (2024). SWOT Adopt a Crossover Field Campaign—Cape Hatteras (MAB-SWOT) Current and Pressure Sensor Equipped Inverted Echo Sounder (CPIES) Technical Report (Version 1). Zenodo. https://doi.org/10.5281/zenodo.12583849

Punongbayan, A. T., **Wang, Y. D.**, Villanoy, C. L., & Yñiguez, A. T. (2022). Connections and clustering of Paralytic Shellfish Toxin events among coastal embayments in an archipelago partly mediated by advection. Harmful Algae, 111, 102147. https://doi.org/10.1016/j.hal.2021.102147

Scholarships and Awards

Stanley W. Watson Student Fellowship Recipient (Woods Hole Oceanographic Institution) 2025
Ralph Rayburn '69 Texas Sea Grant Scholarship (Texas A&M Department of Oceanography) 2022
Society for Underwater Technology Scholarship (Society for Underwater Technology – US) 2022
Phi Kappa Phi Honor Society (Member No. 12613100)

Select Conference Presentations

Wang, Y.D., Andres, M., Chen, S.-Y. Deep-reaching cyclones and their interactions with the Gulf Stream, warm core rings, and the New England seamounts. Ocean 2026, Scotland, United Kingdom. Poster.

Wang, Y.D., Andres, M., Bahr, F., Gawakiewicz, G., Gangopadhyay, A., Porter, N., Jensen, G., Todd, R.E., Lin, Y.T. Characteristics and Environmental Effects of a Warm Core Ring Crossing the New England Seamount Chain. OSM 2022, New Orleans, LA, USA. Poster.

Wang, Y.D., DiMarco, S.F., Kurian, J., Knap, A., Lee, L.L. III. Dense Water Mass Formation and Fate in the Northwestern Gulf of Mexico during the February 2021 Winter Storm Event. AGU 2022 Fall Meeting, Chicago, IL, USA. Oral.

Wang, Y.D., DiMarco, S.F., Knap, A., Walpert, J., Lee, L. III, Hu, X., Chapman, P. (2022, February 24 – March 4). Impact of the 2021 North American winter storms on coastal circulation and hydrographic properties of the northern Gulf of Mexico. OSM 2022, Virtual. Oral.

Outreach and Service

Joint Program Applicant Support & Knowledgebase (JP ASK) Board Member & Mentor 2023 - present
Manage the logistics and advertising of the program and provide supplementary materials to both mentors and mentees; advise and provide support to prospective students in their applications to the MIT-WHOI Joint Program
As an international student on the board, I aim to provide the perspective of an under-represented minority in order to foster an inclusive and accessible environment among mentors and mentees.

Technical Skills

- Human languages: English (native), Tagalog/Filipino (native), Spanish (proficient), French (intermediate)
- Programming languages: MATLAB (advanced), Python (proficient), bash/zsh scripting (proficient), R (intermediate), C (intermediate)
- Proficient in Regional Ocean Modeling System and Delft3D-Flow Hydrodynamic Modelling
- Experienced in handling, maintaining, and training researchers in using oceanographic equipment including but not limited to Acoustic Doppler current profilers, ocean sensors and payloads, weather stations
- Experienced in Liquid Robotics Wave Glider software integration, hardware maintenance and integration, piloting